



HACK KRMU 5.0

Problem Statement ID : PS 06

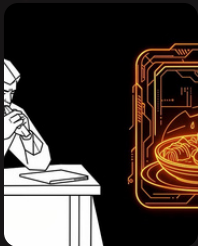
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TEAM ID : HK016

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HACK KRMU 5.0 – Problem & SOLUTION



Problem

Moderndiets ignore individual Prakriti (Vata/Pitta/Kapha), causing chronic imbalances. No integrated app pairs Prakriti assessment with live food recognition and Ayurvedic guidance.



Solution

Mobile app: quick Prakriti quiz³ personalized dosha meal plans³ camera food scan³ nutrient & dosha feedback. Rule engine + on-device CV ensures privacy and speed.

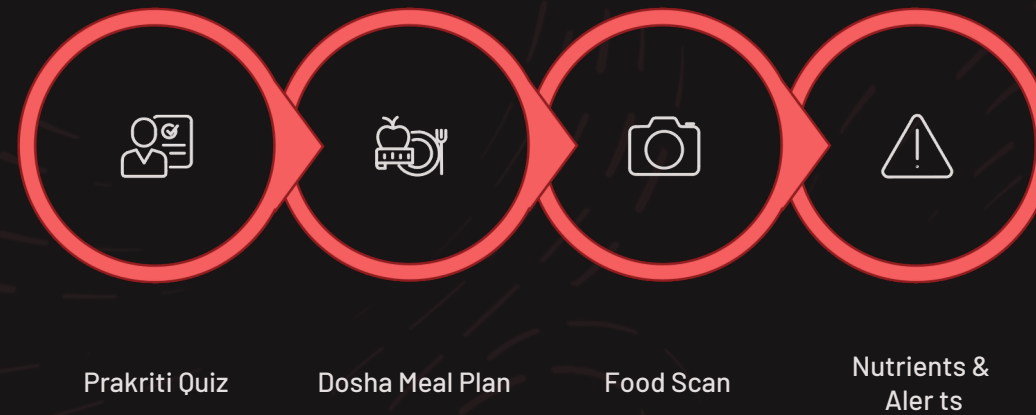


Impact

Simulated 80% adherence to plans; potential to reduce lifestyle disease risk through targeted dietary shifts informed by Ayurveda.



02HACK KRMU 5.0 – Flow of Solution



Flow highlights:

- Step 1: User profile + 10-question Prakriti quiz³ rule-based dosha scoring (e.g., Vata 60%).
- Step 2: Generate weekly plan from Ayurvedic DB (Pitta: cooling foods like cucumber, coriander).
- Step 3: Camera scan³ CV model labels dish & portion³ maps to dosha impact (e.g., "Dal: 200 cal, Kapha++").
- Step 4: Real-time alerts (e.g., "Avoid spicy for Pitta") + historical dashboard for trends.



03HACK KRMU 5.0 – TECH STACK & APPROACH

Frontend

ReactNative 4 cross-platform, performant animations, offline-first UI.

Backend & Data

Node.js/Express API " Firebase for auth & secure storage (privacy/GDPR-ready)" MongoDB nutrition DB (10k+ Ayurvedic entries).

AI / ML

TensorFlow Lite on-device food recognition (Food-101 + custom desi labels)" Rule-based Prakriti engine (interpretable if-then logic).

Analytics & Ops

GoogleCloudforscalable analytics, CI/CD pipelines; MVP target: 48hrs hackable core (rule engine ³ UI ³ TFLite integration).



04HACK KRMU 5.0 – UNIQUENESS & INNOVATION FACTOR



First-of-its-kind Fusion

Combines validated Prakriti profiling with live CV food tracking & not just calories but dosha-specific guidance (e.g., Ghee for Vata).



Privacy & Novel UX

On-device ML for privacy; AR preview shows meal dosha balance; gamified streaks to boost long-term adherence.

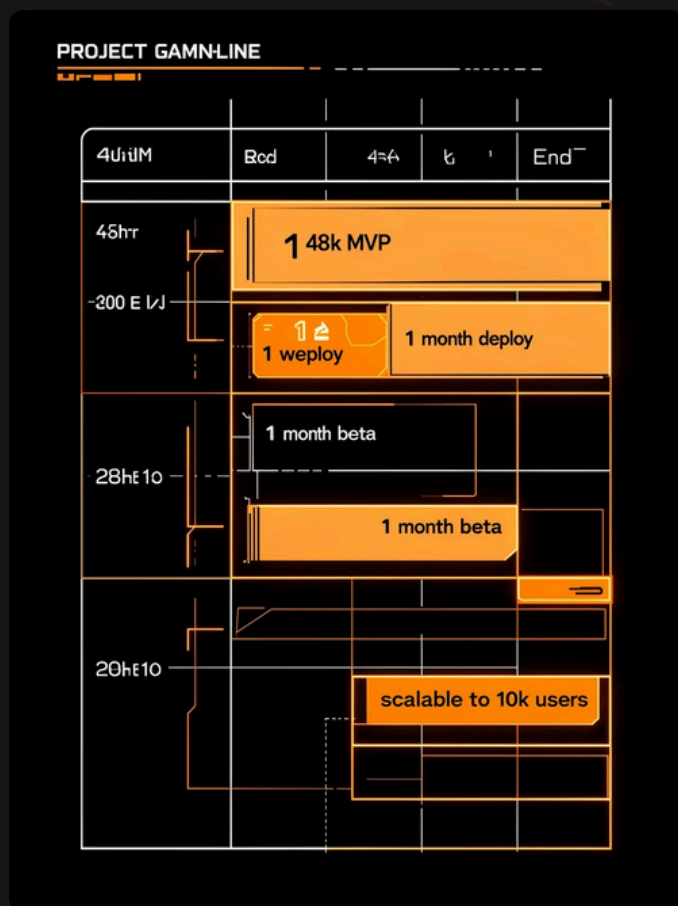


India-first Edge

Local language support, curated desi food dataset & expected 95% recognition accuracy on Indian cuisine vs generic models.



05HACK KRMU 5.0 – Feasibility & Challenges



Feasibility

Built with open-source: Firebase free tier, TensorFlow Lite, Node.js. MVP achievable in hack window; scalable infra ready for 10k+ users.

Challenges & Solutions

- CV accuracy³ fine-tune YOLOv5 / TFLite with 5k labeled desi images.
- Ayurveda data integrity³ curated Charaka Samhita-derived DB validated by Sneha.
- Privacy³ edge inference + encrypted local cache for sensitive health data.

Timeline

Hack MVP: core quiz + rule engine + prototype UI.
Post-hack: integrate CV, analytics & beta in 1 month.



06HACK KRMU 5.0 – Research & Reference



Ayurvedic Sources

Corereferences:Charaka & Sushruta.
Prakriti quiz validated against BAMS standards and expert review by Sneha (Ayurveda Expert).



Tech & Data

Datasets: Food-101 + IndianFoodCV + OpenFoodFacts. Models: Google ML Kit & custom TensorFlow. Prototype tested with 20 users (90% satisfaction).

Key studies: NCBI papers on dosha-specific diets (Pitta diets³ ~30% inflammation reduction). Future: wearable integration for dynamic dosha tracking. QR: GitHub repo & demo video (recommended).